Responsible Software - CheatSheet

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INTRO. Responsible Software. Software engineered in a responsible *(ethical)* way.
1. With a goal to do good. 2. While preventing avoidable negative impacts.
3. Taking people, other systems, social structures and our planet into account.
Active Responsibility. anticipating/preventing negative impacts before occurring.
Ethics. Principles/standards that guide human behavior in various contexts.
Metaethics. "what does it mean for an action to be right ?"
Normative. "how should people act in general (moral principles)?"
Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have *ethical* implications.
Ethical Issue. any situation that may compromise respect of ethical value/principle.
Ethical Agency. capacity to take actions that align with ethical values and principles.
Ethical Sensitivty. ability to
- recognize ethical issues (awareness)
- identify their impact: who, what

Ethical Decision Making.
1. **Recognize**(knowledge+experience+habits)
2. **Evaluate**(cognitive+reason+affective) 3. **Act**.
Ethical Lenses.
1. **Desirability.** product corresponds to what people want.
2. **Viability.** product generates profits for a business.
3. **Feasibility.** product can be created with existing or new technology.
Fourth criterion (Isaac et al.). add **Ethicality.**

Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. *None of the options override the other, wrong-wrong choice.*
Moral side.
Emotions should be used explicitely/intentionally for <u>responsible</u> design (Roeser). In response to moral issues or motivating immoral behavior.
Cognitive side (Normative Ethics).
- **Utilitarianism(Consequences).** choose the greatest good for greatest number
- **Deontology(Action).** do right according to universal principles
- **Virtue(Person).** act as a virtuous person
- **Care(Relationships).** mutual responsibility/care for each other

Trolley problem. thought experiment, **choice**: do nothing → runaway trolley kill people, <u>or</u> intervening → fewer people die. tests which moral principles justify action vs. inaction.

Ethical Decision Frameworks.
1. Identify the ethical issues
2. Get the facts
3. Evaluate alternative actions with *several ethical theories*
4. Choose an option and "test" it
5. Implement your decision and reflect on the outcome.

Example questions for brainstorming
- Who will use your product on a daily basis?
- Who is in geographical proximity with the users of your product?
- Who is in social proximity with the users of your product?
- Who is involved in the maintenance of your product?
- Who provides services that are needed to run your product?
- Who provides funding?
- Which communities are active in the application domain?

Stakeholder Individuals/groups/organizations/institutions/societies that might reasonably be affected by technology. **Direct Stakeholder.** *in contact* with the product
Indirect Stakeholder. affected through *consequences of the usage* of the software
<u>**Stakeholder analysis.**</u> Identifying every entity that can be affected by software positively/negatively. 1. Find all stakeholders possible using questions.
2. Use Direct vs Indirect Chart to find new stakeholders.
Additional Methodologies: *Ethix. add unintended stakeholders, Mitchell. add attributes to stakeholders, SAS. use different chart with scale for impact from product and influence on the product.*

<u>**Ethical Speculation.**</u> How to think, not what to think - anticipate potential impacts on stakeholders.
Create a fictional story in two parts. (*done at the early design phase*)
I. Dark and Pessimistic
1. Imagine a character that would be negatively impacted by the software
2. Write a pitch that summarizes their story.
3. Find a title.
II. Positive outcome
1. Identify 1 or 2 main ethical issues represented in your story.
2. Describe the immediate and future consequences.
3. Imagine a happy ending for character.

<u>**SAFETY II.**</u> **Levels of analysis.** Macro (societies), Meso (groups/communities), Micro (individuals)
Internet Trafic. 99% submarine cables: 1.4M km, 4.7B users, 1.2 ZettaBytes (1.2 trillion GB).
Internet Benefits (+). Information access/dissemination, Communication, Social relations/organization, Identity/psychological development, Learning/cognitive development, Production/commerce.
Internet Harms (-). Privacy boundary violations, Social relationship harm, Community damage, Addiction, Cognitive impairment, Information overload, Knowledge/belief distortion, Democratic threats.
Tension. Same features enable both benefits/harms → *social connection enables social harm, information access causes overload, identity spaces blur privacy boundaries.*
From micro to macro.
Types of impacts: lots of individuals, Social structures/institutions, Different for different groups.
Areas: Markets/economy, Knowledge/worldviews, health/well-being, Politics/citizenship, Environment.
⚠**controversy** Complex measurements, causality hard to establish.

Disinformation. false info. with intent to deceive.
Misinformation. false info. without intent to deceive
Malinformation. true info. used with harmful intent.
fake news → false information, with or without intent.
Global risks. impact on electoral processes *(Brexit, U.S. elections)*, growth of distrust (government/media sources), risks of repression. *socio-technical phenomenon*
Cognitive effects. how we think.
Socio-affective effects. interaction with people and emotions.
System 1. fast/emotional/ activated first as quick response.
System 2. slow/logical/conscious analysis.
Illusory truth effect. repetition increases perceived truthfulness of claims by enhancing familiarity and intuitive/shortcut-based thinking.
Socio-affective factors
- **Source cues.** perceived credibility (powerful/attractive sources more likely to be believed).
- **False consensus.** overestimating how many people agree *(climate change denial.* ~ 6%) *in reality, perceived* ~ 50%).
- **Emotions.** strong emotions content/receiver emotional stae increases susceptibility.
- **Identity resonance.** alignment with personal identity/values increases acceptance.

Continued influence effect: misinformation continues to influence beliefs even after correction is received and accepted as true.
Increased use of digital media shows greater exposure to misinformation, increased political polarization, decreased trust, and increased hate. *empirical results remain debated and sometimes exaggerated*

Publication. publicly availabling content, *financial/ political/ coordinated strategic* incentives.
Human publishers. Influential figures/ordinary users publishing for visibility/revenue, often using *clickbait*/participating in organized campaigns *(coordinated troll operations)*.
Social bots. Automated accounts publishing at scale (**1–15%**), they generate *disproportionately large share* of misinformation.
Propagation. content spreading through *reposting/ liking/ algorithmic amplification.*
Human propagation. content reshared by real/fake accounts. engagement inflated by paid workers, while moderation reduces resharing *(account bans lowering repost rates).*
Genuine resharing factors. Resharing driven by *inattention, emotional reactions*, and *socio-affective motives* such as signaling group membership or provoking reactions *(outrage).*
Social bots in propagation. Automated accounts rapidly *repost* and *target influential or opposing users* to maximize visibility and conflict.
Recommender systems. *select, rank, display* content, optimize for *engagement* (*popularity bias*).

Ethics Canvas. Structure stakeholder, impact, and mitigation info systematically.
When to use? Early software development *Brainstorm, visualize, mitigate ethical impacts)*
1. Identify Stakeholders: Who is affected?
2. Ethical Impacts: What are the impacts on them?
3. Actions: How to address these impacts?

<u>**Causal loop diagram.**</u> to explain complex systems/improve performance/solve issues
focuses on specific issues *(amount of fake news, investment in an auto-detection system)*
Causal links are labeled with a +/− longer delays are represented with thicker arrows.

<u>**Edge Cases.**</u> Foresee consequences/identify opportunities for improvement analyzing extreme scenarios.
When to Use: At all stages,design/major updates/user base growth/audience shifts to address evolving societal impacts.
– Global Reach: Impact of multicultural/lingual use.
– Mass Adoption: Scalability/infrastructure under high loads.
– Longevity: Long-term use/societal changes/evolving norms
System. collection of organized parts that interact as a whole to reach a state/purpose; modeling simplifies complexity/identify patterns.

<u>**SAFETY I.**</u> **Security.** Prevent the environment from affecting negatively the system. ⚠*threat* | *Vulnerability scanning, ethical hacking, security audit, code review*
Safety. Prevent the system from affecting negatively the environment. ⚠*hazard* | *Verification and testing, failure modes analysis, Redundancy*
Common challenge - human factor → consider software as **socio-technical system**
Internet/social media; **human behavior/technical systems constanly influence each other.**

Malfunction. software doesn't function/perform normally/as specified.
- Due to unintentional defect/fault/bug→critical consequences(⚠Life/Mission-critical systems)
Misuse/Abuse. attack, diversion, use of software for *nefarious purposes.*
Bad/threat actors. *malicious intent, authorized/unauthorized users*→direct/indirect impact.
Unintended use. use of software in a way not originally intended/not (officially) supported.
⊕ creates opportunities for innovation *use camera to scan documents* ⊖ *use llm to obtain factual information*
Intended use software used as intended. difficult to anticipate impact due to social ⚠scale/time

User-Generated Content (UGC). any content posted by users online *images/videos/texts*
- Published publicly and (usually) not produced by professional/institution *wikipedia*
⚠**Toxic content.** material harmful/offensive can have negative impact on individuals/communities
Types of impacts? Physical/Financial/Emotional/Relational.
Hate and Harassment.
Violence threats, Identity misrepresentation
Self-inflicted harm. Eating disorder promotion
Ideological harm.
Extremism/terrorism, Misinformation
Exploitation. Sexual abuse, Scams
Hate speech
- Discriminatory/pejorative of individual/group
- calls out real/perceived "identity factors" of individual/group
- conveyed through any form of (images/memes/…)

Central dilemma. protecting users from toxic content while preserving free speech rights
To decide?
- State (responsible for fulfilling rights),
- Platforms (private actors with business interests),
- People (responsible for the content they publish.
Free speech. right to express opinions and to seek, receive, share information/ideas, in any format offline/online, subject to limitations that respect the rights of others and prohibit incitement to discrimination or violence, pillar of democracy.

UGC platforms provide channels for hate speech. Easily produced, shared, anonymous, global/diverse audience, realtime, amplified by content recommendation algorithms

Predicted \ Actual	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)

Content moderation:
False negative: toxic content that escapes moderation.
False positive: wrongful censorship, restricting free speech.

TP. Model predicts positive and is correct
Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
Positive Predictive Value (PPV) = $\frac{TP}{TP+FP}$
Negative Predictive Value (NPV) = $\frac{TN}{TN+FN}$
False Discovery Rate (FDR) = 1 − PPV = $\frac{FP}{TP+FP}$
False Omission Rate (FOR) = 1 − NPV = $\frac{FN}{TN+FN}$
False Positive Rate (FPR) = $\frac{FP}{FP+TN}$
False Negative Rate (FNR) = $\frac{FN}{FN+TP}$

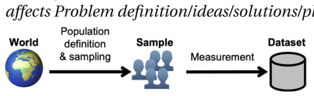
STRIDE Threat Modelling. Identify computer security threats across six categories.
When to Use: At any project stage, complementing bad actors modelling.
– S: Spoofing – Impersonation attacks.
– T: Tampering – Unauthorized modifications.
– R: Repudiation – Denial of actions without proof.
– I: Information Disclosure – Data leaks.
– D: Denial of Service – Service unavailability.
– E: Elevation of Privilege – Gaining unauthorized access

<u>**Bad Actors Modelling**</u> Minimize vulnerabilities via risk assessment and mitigation.
When to Use: Design phase, prior to updates, audits, incident responses, and improvements.
What is a Bad Actor? A person/group exploiting a system for personal goals.
Motivations (might overlap):
- Money: Data theft, ransomware, sabotage by competitors.
- Politics: Misinformation/strategic sabotage/infrastructure attacks.
- Entertainment: Attacks on random systems, misinfo. spreading.
- Self-interest: harassing messages/commun. monitoring/location tracking.
- Ideas: Discrediting groups, data leaks, website defacement.

Steps: 1. Identify possible malicious users and their motivations.
2. Assess potential consequences for the system and users.
3. Develop measures to prevent these negative outcomes.

<u>**Harm Modelling**</u> anticipate/address potential harms users/stakeholders may encounter.
When to Use: At every stage of project, for every feature, until harm potentials are satisfactory.
Evaluate: 1. The category of use. 2. the types of harms. 3. the magnitude of harm
Categories of Use:
Harm from…
Intended Use: correct usage.
Unintended Use: unexpected usage.
Misuse/Abuse: malicious usage.
Malfunction: system failures.
Types of Harm:
Humans: Physical, emotional injury.
Allocation of Resources: Opportunity, economic losses.
Human Rights: Dignity, liberty, privacy losses; environmental impact.
Social Systems: Manipulation, social detriment.
Magnitude of Harm:
Severity: Impact on individual/group well-being.
Scale: How broadly the harm affects populations.
Probability: Likelihood of harm occurring.
Frequency: How often the harm might occur.

<u>**FAIRNESS I.**</u> **Data in Software.** people described by **attributes** (stored in databases/files), can be *sensitive.*
⇒ *ethical implications* such as *confidentiality, privacy, and fairness.* data is an *imperfect representation of real world*, must be *simplified* to fit context.
Data protection law *(GDPR)* - **Personal** data: reveals info about *identity*, directly/indirectly.
- **Special category** data: banned by default. *racial/ethnic origin, pol. opinions, rel. beliefs, health data.*
Nondiscrimination Law forbids direct/indirect discrimination, when **apparently** *neutral* practice puts *specific groups* at disadvantage. *(Protected Group/Class: Category qualifying for special protection by law/policy)*
Proxy Attributes. Variables that **correlate** w/ sensitive attributes.
Risks & Outcomes. can lead to **indirect discrimination** *(harder to detect).*
- Software has discriminatory outcomes even in absence of sensitive attributes in data.
(high school quality) - correlates w/ income/ethnicity → becomes proxy.
Fairness. Equality *opportunities*; operationalized by treating people same. Scope matters: groups not protected law still face ethical harm.
Unfairness. Discrimination, stereotyping, decision errors based irrelevant factors; causing negative impacts via biased software/processes *(women get lower credit limits despite higher income).*
Bias. Systematic deviation from standard. context-dependent, not always harmful, possible ethical issues.
affects Problem definition/ideas/solutions/planning; shaped by external factors (info/org. culture)



1. **Pre-existing bias.** Bias already present world, captured in data *(systemic bias).*
2. **Representation bias.** Data inaccurately reflects target population/subgroup *(Sampling: over/under-representation. Population: sampled pop. differs actual users).*
3. **Measurement bias.** Methods yield inaccurate values, oversimplify reality, differ accuracy across groups *(blood-oxygen devices more accurate white people vs darker skin due melanin).*
4. **Cognitive bias.** Human (often implicit) deviations rational judgment; affects info processing/interpretation; enters via users/designers.
Stereotypes. Generalized beliefs about group *(implicit/explicit; pos/neg/neutral)*; ignore individual differences, reinforced by design assumptions.
Group effects. groups with same opinions/preferences → reinforce confirmation bias; *(devil's advocate strategy)* doesn't work → groups with heterogeneous preferences more balanced
→ group-think: phenomena where alternatives are avoided, pseudo-consensus
Sunk-cost fallacy. continuing course of action just due to past investment → prevents alternatives→product with design flaws → **Technical debt.** future costs to fix, **Ethical debt.** ethical consequences/harms

<u>**Inclusive Design.**</u> design software that benefits the widest range of users by addressing diverse abilities, languages, cultures, and differences.
When to Use: Early in soft. development phases.
1. Identify needed capabilities for use.
2. Identify and label "Non-Average Users" (NAUs).
3. Add unaddressed capabilities/new user categories.
4. Develop and report concrete inclusive solutions.
Aim: Ensure artifacts match the needs of diverse individuals beyond average users.
Capabilities. perception(vision,hearing,…), thinking, action(dexterity…)

<u>**Analyzing Values.**</u> Identify/analyze/evaluate values embedded in technology and resolve value-based tensions.
What are Values? principles/desirable goals for design, based on Schwartz's value theory.
1. Identify values/stakeholder values/artifacts values.
2. Analyze value-based benefits (supporting stakeholder values)/harms (opposing stakeholder values).
3. Map value tensions:
* Within the same value (e.g., benefit vs. harm).
* Between different values (interpret context to resolve).

FAIRNESS II. ML. techniques inferring patterns from data, generalizable to unseen data.
Process. Iterate to find parameters of function that best represent pattern in data.
Result. Function + function parameters.

Training. Run learning algorithm to find parameters → **Inference.** compute results on previously unseen data.
Regression. Output continuous values *(house price from size)*.
Classification. Output discrete values *(wine vs beer from alcohol level, color)*.
Supervised. Output labels given. *(labels created by humans; picture descriptors)*
Unsupervised. No output labels *(clustering, dimensionality reduction)*.
Data dependence. ML depends on data → sensitive to biases; in supervised, labels human-made → not all issues attributable to ML alone.
Pipeline. Problem definition → data collect + preparation → model dev + model eval + model deploy.
Model dev. Choose model type, objective function, hyper-parameter tuning for performance → unfairness risk when optimizing accuracy only.
Aggregation bias. Combine groups with different patterns → model performs poorly for some groups, or mainly for majority.
Evaluation bias. Benchmark not representative of served population → *(Gender Shades: intersectional bias across gender and race)*.
Deployment bias. Mismatch between intended problem and actual use → socio-technical bias.
Fairness metrics. Mathematical measures of fairness notions *(criticised for oversimplification; focus outcome)*.
Error parity. Error rates independent of group. Contains:
1. **Error rate balance.** Equality of error rates compared to actual values→FPR, FNR across groups.
2. **Conditional use accuracy equality.** Equality of predictive values compared to predicted values→PPV,NPV across groups. 3. **Equal accuracy.** Equality of accuracy across groups.
Demographic parity(Equal acceptance rate). Positive prediction rate independent of group *(focus outcome)*: $\frac{TP+FP}{Total}$
Disparate impact ratio. $\frac{\text{positive prediction rate (group A)}}{\text{positive prediction rate (group B)}}$ should be 1.
Impossibility result. Cannot satisfy all fairness metrics simultaneously.
Group fairness criteria (GFC).
Quantify equality between groups; bundle people treated differently → aim for group-level equality.
→ Mathematical impossibilities among parity measures, incl. with demographic parity.
→ Unfair world + imperfect classifiers → not all group fairness criteria hold simultaneously → choose priority notion of fairness.
Issues of GFC.
May miss unfairness at intersection of multiple groups.
→ Need intersectional group fairness → many small groups.
→ No individual differences within groups → need individual fairness.
Individual fairness. Similar individuals should receive similar decisions; requires similarity metric, tricky.
Causal analysis of fairness. Focus causes of unfairness and causal pathways from sensitive attributes to outcomes → find proxies via causal graphs.
Improve fairness. Either modify data or modify model.
Modifying the data. *(pre-processing)* If data biased, learned patterns biased. **To reduce bias**, *Fairness through unawareness* removes sensitive attributes from dataset, but reduces accuracy. If proxies exist, bias may persist. Other ways: weighting, resampling (ensure better representation of concerned groups), or transformation (transform data with function preserving data while hiding bias).
Acting on the model. At training time: *(in-processing)* optimise accuracy and fairness together, add fairness metrics; or *(post-processing)* directly apply transformations to output, independent from model.
To improve fairness of model, almost always face dilemmas:
→ **Leveling down.** More fairness for some might compromise fairness to others.
→ **Fairness–accuracy.** More fairness may often compromise accuracy.
→ **Fairness–privacy.** Access to sensitive attributes, increasing privacy may compromise fairness.

Datasheet for dataset. Fill form with technical details/explanations about data, collection process, processing, intended usage→for devs: avoid publishing ill data; for users: ensure data cleanliness, better understand dataset.
Model card. Same idea but for ML → technical info, intended use, performance, metrics, data for training/evaluation, quantitative analyses, ethical reviews, usage recommendations.

AI Ethics card (IDEO). Guide ethically responsible, culturally considerate design with data.
Principles: data ≠ truth. don't presume AI durability, respect privacy/collective good, unintended AI consequences → design opportunities. **Goal: humanize data *(people behind data)*.** Method: find/read documentation, raw data, materials; select variables/columns; pick few; inspect their data; focus on one variable (select people random or by specific characteristics); write conclusion in terms of ethical impact and risks.

Sustainability I. Capacity to endure across full lifespan (launch to end-of-life); linked to maintenance/evolution. **Three pillars view.** Economy, environment, society (treated as substitutable).
Strong Sustainability. Recognizes the interdependence of economic, social, and environmental systems, emphasizing the irreplaceability of natural resources. **Technical Sustainability.** Focuses on software durability/its maintenance and evolution over time.
Nested view. Economy part of society; society part of environment → not substitutable.
Green by software. Software supporting sustainability goals of other systems.
Green in software. Software itself designed to be sustainable (focus).
When? - Development. Requirements, design, implementation, testing, distribution, deployment.
- **Usage.** Operation, maintenance, evolution.
- **End of life.** Decommissioning, disposal.

What. Software cannot run alone → requires hardware and network together.
Beyond carbon. Carbon footprint matters *(ICT sector: 1×10^{12} kg CO_2, more than France or Australia)*, but assessment should cover broader planetary boundaries *(9 critical processes)*.

Global warming. Increase in multi-decade global mean surface temperature above industrial level (1900): +1.1°C (2011–2020), +1.45°C (2023). due to human activity/greenhouse gases.
Greenhouse gases (GHG). Atmospheric gases trapping heat in Earth's atmosphere (natural phenomenon); increased GHG concentrations raise temperature. Main contributors: CO_2, CH_4, N_2O.
Software and electricity. Software needs electricity; electricity generation emits GHG → software has CO_2 emissions depending on energy consumption and carbon intensity of electricity production.
Carbon intensity examples. *(coal: 820 g CO_2e/kWh), (gas: 490 g CO_2e/kWh), (solar: 41 g CO_2e/kWh)*.
Location. Electricity origin depends on location; carbon intensity computed from average energy mix (fossil fuels, nuclear, renewables). **Time.** Electricity mix varies over time; renewables not produced continuously.
Optimizing software energy. Shift execution in time and location to exploit cleaner electricity mixes.
GHG Protocol for companies.
- **Scope 1.** Direct emissions from sources owned/controlled by company.
- **Scope 2.** Indirect emissions from purchased/consumed energy.
- **Scope 3.** Other indirect emissions across value chain (suppliers, customers/consumers).

Programming languages and efficiency. Languages differ in efficiency; faster ≠ more energy-efficient.
Main energy drivers. Majority from CPU: algorithmic complexity, loops, threads, data structures; also RAM, disk transfers, and I/O transfers matter.
Hardware and OS role. Idle power draw exists; OS/hardware power-saving systems (low power modes) affect consumption.

Data centers. Virtualization lets one physical server run multiple workloads; consolidation can reduce energy if optimized. Cooling critical; data centers powered continuously. Typical split: servers 50%, cooling 40%, remainder losses/other.
Power Usage Effectiveness. $\text{PUE} = \frac{\text{total data center energy}}{\text{IT equipment energy}}$. Ideal 1, typical ≈ 1.9.
Distributed software. Multiple components deployed across infrastructure → distributed energy consumption (including user devices) + network. **Usage-stage split.** 50% user devices, 25% data centers, 30% network.
Network accounting. Estimation hard (constantly changing; boundary definitions vary). *(60 Wh/GB; phone charging: 22 Wh)* ⚠Energy efficiency vs time, cost, performance, maintenance.
Rebound effect (Jevons' paradox). Efficiency gains reduce service cost → increased consumption; existence widely accepted, magnitude debated. **Direct rebound.** Consumers use more of the same service.
Indirect rebound. Savings spent on other energy-consuming goods/services.

Decision Matrix. Make decisions by evaluating and prioritizing options based on multiple criteria.
When to Use: To narrow down a list of options to one choice. When decisions must consider several criteria.
1. Brainstorm and identify evaluation criteria.
2. Refine criteria and assign relative weights *(distribute 10 points)*.
3. Create an L-shaped matrix with criteria and weights along one axis, and options along the other.
4. Rate each option consistently *(1–5 scale)*.
5. Multiply weights by ratings, sum results, and select the option with the highest total score.

SUSTAINABILITY II. FLOPS. Floating point operations → growing 4×/year since 2010.
Examples of ML training footprint. GPT-3 (175B param) → 1300 t CO_2e; BLOOM (176B param) → 433 t CO_2e.
Carbon budget. ≈ 2 t CO_2e/person/year if we want to keep global warming under control.

PUE (Where). Data-center efficiency overhead (cooling + power delivery losses).
CI (When). Grid carbon per kWh at that time (electricity mix).
$C_{\textbf{training}}$. training time in hours and power consumption in W and power usage.
$C_{\text{training}} = t \times (P_{\text{CPUs}} + P_{\text{GPUs}} + P_{\text{ME}}) \times \text{PUE} \times \text{CI}$.
Quantity training. Need train model multiple times *(LISA: 200× footprint of 1 train; BLOOM only 2.6)*.
$C_{\text{inference}} = N_{\text{queries}} \times E_{\text{inference}} \times \text{PUE} \times \text{CI}$.
Energy per query. BLOOM 0.004 kWh/query and 0.28 kWh in idle.
Generative tasks most energy-intensive; image tasks > text *(can be as much as charging one phone)*. Multipurpose model > task specific model.
$C_{\textbf{ML}} = C_{\textbf{dev}} + C_{\textbf{inference}}$ $C_{\text{inference}} \approx 60\%$ of total. C_{dev} can be large (one-time), $C_{\text{inference}}$ scales with usage.
Embodied emissions. Greenhouse gas emission associated with production, transport, end of life → carbon already embedded into product *(36% embodied, 64% usage)*. One-shot cost → attribute share based on exec time and used resources.

Life Cycle Assessment. Methodology to evaluate range of environmental impact across all life cycle phases → impact on all, not just carbon footprint *(water, metals, ...)* → long process.
Product Carbon Footprint (PCF). LCA reduced to Global Warming Potential and carbon.
→ Estimate with databases + math modeling.
→ Highest embodied emissions: integrated circuits (CPU, RAM, ...) + displays; mainly manufacturing; depends on component type, perf, dimensions.
→ Take embodied emissions, timeshare *(exec time / hardware lifespan)*, resource share *(exec resources / hardware resources)*.
PCFs. Embodied emissions + usage emissions. Fix usage: better hardware, but better hardware has more embodied emissions → check soft/hard-ware dependencies. Lighter software reduces both; reducing embodied extends lifespan and utilization. **Water withdrawal.** Total amount of water taken from a source.
Water footprint (= water consumption). Difference between water taken and water returned *(evaporated, incorporated, consumed, or returned at different location/time)*.
Blue water footprint. Freshwater consumption *(surface + groundwater)*, incl. potable water *(most competition between technology and humans)*.

Green water footprint. Rainwater consumption *(plants/soil)*.
Gray water footprint. Water consumed to dilute pollutants so water meets quality standards.
Energy tech footprints. Large footprint: biomass, hydropower. Low footprint: solar photovoltaics, wind.
EU electricity (2015). Total water consumption for electricity production: 1.54×10^{12} L.
Thermal power plant. Heat-to-electricity; needs water ⇒ high consumption; depends on cooling type; similar issue in data centers; potable water can be used.
Water usage effectiveness (WUE). $\text{WUE} = \frac{\text{Annual water usage}}{\text{Energy of IT equipment}}$ *(electricity treated as direct consumption)*.
Cooling. Sensitive to temperature/humidity; varies with climate, location, season, weather conditions ⇒ water footprint varies; considered indirect consumption.
In energy production, often a water vs. energy(carbon) tradeoff.
WUE Source. *(cooling direct, electricity indirect)*.
$$\text{WUE}_{\text{source}} = \frac{\text{Annual site (cooling) water usage} + \text{Annual source-energy (electricity) water usage}}{\text{Energy of IT equipment}}$$
Water cycle. Water changes state and location; never lost, but availability in time and place crucial.

Ethical Decision Making. explore ethical aspects of decisions/evaluate options using multiple ethical lenses.
What is a Lens? A normative ethical theory used to evaluate decision options.
The 6 Lenses:
– Rights: Respect stakeholders' rights.
Which option best respects everyone's rights?
– Justice: Treat people fairly and equitably.
Which option ensures fairness?
– Utilitarian: Maximize good, minimize harm.
Which option benefits the most stakeholders?
– Common Good: Serve the community, especially the vulnerable.
Which option serves the community best?
– Virtue: Strive to be an ideal person.
What kind of person am I if I choose this?
– Care Ethics: Prioritize relationships and feelings.
Which option considers all stakeholders' concerns

EMPOWERMENT I. Choice architecture. Context/environment where decision is made.
Nudge. Aspect of choice architecture altering people's behavior predictably without forbidding options or changing economic incentives → intentionally bias decisions.
→ Applied in gov, health care, promote greener behavior, info security, privacy.
Social nudges. Reference to users of how other users behave → create social norm.
Default. Choice assigned in absence of active decision, without removing choice
→ auto subscription renewal, newsletters → significant effect on chosen option.
- **Opt-in.** Checkbox unticked → user actively ticks.
- **Opt-out.** Checkbox ticked → user actively unticks → doubled organ donors *(40% → 80%)*.
→ Threats to 3 aspects of people's autonomy.
Factuality errors. Factually incorrect or fabricated content.
Faithfulness errors. Content inconsistent with instructions/input, or without internal logic.
Automation bias. More confidence in automated systems than other sources (including self)
→ over-rely on automated systems and recommendation systems even if output incorrect, even if own judgment would be correct.
Autonomy. Freedom of choice from external pressure hard to resist; agency capacity to reason/reflect to make choices; self-constitution; identity, values, authorship on life.
→ Nudges can touch autonomy, often low transparency; consider whether design increases individual/societal welfare; can cause infantilization, distrust.
Automated systems can act on. Information acquisition, information analysis, decision selection, decision implementation
→ ChatGPT produces plausible text, no warranty it is correct.
→ Called "hallucinations"/errors; in artificial neural networks; can cause harm but also be valuable; error rate hard to know.